Empirical Plots

Setup

```r
library(here)
library(arrow)
library(tidyverse)
library(data.table)
library(Hmisc)
```

```r
set.seed(1234)
df_weekly <- read_parquet(here("data", "hosp_long_weeks.parquet"))
df_daily <- read_parquet(here("data", "hosp_long_daily.parquet"))
```

Create death carried forward dataset

Weekly

```r
df_state_5 <-
  df_weekly |>
  group_by(pat_id) |>
  mutate(
    y = as.integer(y),
    yprev = as.integer(yprev),
    death_time = ifelse(any(y == 5), week[y == 5], NA_real_)
  ) |>
  filter(y == 5) |>
  complete(week = death_time:26,
           fill=list(y=5L, yprev=5L, gap=1)) |>
  arrange(pat_id, week) |>
  fill(everything())

df_state_1_to_4 <-
  df_weekly |>
  mutate(yprev = as.integer(yprev)) |>
  filter(y != 5) |>
  arrange(pat_id, week)

df_forward_weekly <-
  bind_rows(df_state_1_to_4, df_state_5) |>
  mutate(y = as.ordered(y),
         yprev = as.factor(yprev)) |>
  arrange(pat_id, week)
```

Daily

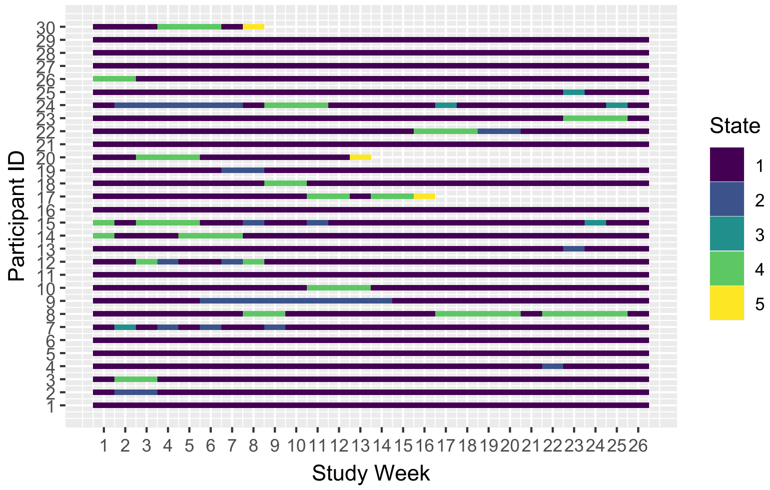
```r
df_state_5 <-
  df_daily |>
  group_by(pat_id) |>
```

```
  mutate(
    y = as.integer(y),
    yprev = as.integer(yprev),
    death_time = ifelse(any(y == 5), day[y == 5], NA_real_)
  ) |>
  filter(y == 5) |>
  complete(day = death_time:182, fill = list(y = 5L, yprev = 5L)) |>
  arrange(pat_id, day) |>
  fill(everything())

df_state_1_to_4 <-
  df_daily |>
  mutate(y = as.integer(y), yprev = as.integer(yprev)) |>
  filter(y != 5) |>
  arrange(pat_id, day)

df_forward_daily <-
  bind_rows(df_state_1_to_4, df_state_5) |>
  mutate(y = as.ordered(y), yprev = as.factor(yprev)) |>
  arrange(pat_id, day)
```

Weekly Plots

Tile plot

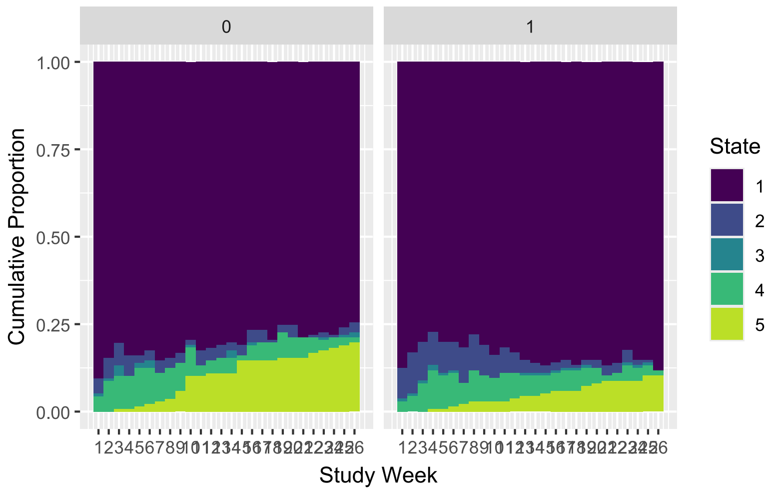
```
ids <- sample(unique(df_weekly$pat_id), 30)

df_weekly |>
  filter(pat_id %in% ids) |>
  mutate(pat_id = as.integer(factor(pat_id))) |>
  mutate(y = as.ordered(y)) |>
  ggplot() +
  aes(y=pat_id, x=week) +
  geom_tile(mapping = aes(fill = y),
            width=1, height=0.4) +
  scale_fill_viridis_d() +
  scale_x_continuous(breaks = 1:26, limits = c(0.5, 26.5)) +
  scale_y_continuous(breaks = 1:30) +
  labs(x = "Study Week",
       y = "Participant ID",
       fill = "State")
```




```r
ggsave(filename = here("img", "fig-tile-hosp-weekly.svg"), height = 10, width = 10)
```

Empirical SOPs

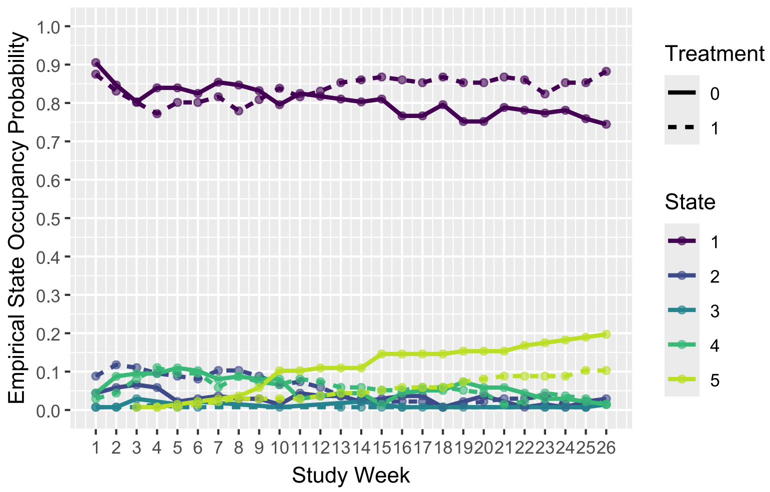
```r
df_forward_weekly |>
      mutate(y = as.factor(y)) |>
      group_by(week, y, tx) |>
      count() |>
      group_by(week, tx) |>
      mutate(n = n / sum(n)) |>
      ggplot() +
      aes(x = week, y = n, fill=y) +
      geom_col(position="fill", width = 1) +
      scale_x_continuous(limits = c(0.5, 26.5),
                         breaks = 1:26) +
      scale_y_continuous(breaks = seq(0, 1, 0.25)) +
      scale_fill_viridis_d(end = 0.9) +
      coord_cartesian(ylim = c(0, 1)) +
      facet_wrap(~tx, nrow=1) +
      labs(x = "Study Week",
           y = "Cumulative Proportion",
           fill = "State")
```



```
ggsave(filename = here("img", "fig-emp-sop-hosp-weekly.svg"), height = 10, width =
12)
```

SOP lines plot

```
df_forward_weekly |>
  mutate(y = as.factor(y)) |>
  group_by(week, y, tx) |>
  count() |>
  group_by(week, tx) |>
  mutate(n = n / sum(n)) |>
  ggplot() +
  aes(x = week, y = n, color=y, linetype = tx) +
  geom_line(linewidth = 1) +
  geom_point(alpha=0.6, size=1.5) +
  scale_x_continuous(breaks = 1:26) +
  #scale_color_brewer(palette = "Set2") +
  scale_color_viridis_d(end = 0.9) +
  scale_y_continuous(breaks = seq(0, 1, by=0.1)) +
  coord_cartesian(ylim = c(0,1)) +
  labs(x = "Study Week",
       y = "Empirical State Occupancy Probability",
       color = "State",
       linetype = "Treatment",
       #title = "(B)")
  )
```



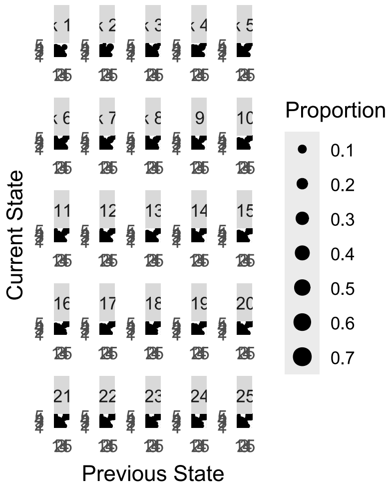

```
ggsave(filename = here("img", "fig-emp-sop-line-hosp-weekly.svg"), height = 10, width =
12)
```

Transitions

```
fu <- 26
# Create the transition data.frame for week = x
get_transitions_df <- function(df, x){
```

```r
  subset <- df[df$week == x, ]
  subtable <- table(subset$y, subset$yprev, dnn = c("y", "yprev")) |>
  as.data.table()

  subtable[, week := x]
  return(subtable)
}

# Get the transition data frame for each week and combine
# into a single data frame
df_transitions <-
  map(2:fu, ~get_transitions_df(df_forward_weekly, .x)) |>
  rbindlist()

# Compute the proportion of transitions relative to the number of
# transitions on each week
df_transitions <-
  df_transitions |>
    mutate(total = sum(N),
           p = N / total, .by=week)
```

```r
FONT_SIZE <- 10
df_transitions |>
    #filter(week%%2 == 0) |>
    mutate(yprev = as.factor(yprev), y = as.factor(y)) |>
    filter(p != 0) |>
    ggplot() +
    aes(x = yprev, y = y, size = p) +
    geom_point() +
    scale_size_area(breaks = seq(0, 1, by = 0.1), max_size = 4) +
    facet_wrap(~week,
               nrow = 5,
               scales = "free",
               labeller = labeller(week = \(x) glue::glue("week {as.integer(x)-1} to
{as.integer(x)}"))) +
    scale_x_discrete(drop=FALSE) +
    scale_y_discrete(drop=FALSE) +
    theme(axis.ticks = element_blank()) +
    theme(aspect.ratio = 0.9) +
    labs(x = "Previous State",
         y = "Current State",
         size = "Proportion")
```



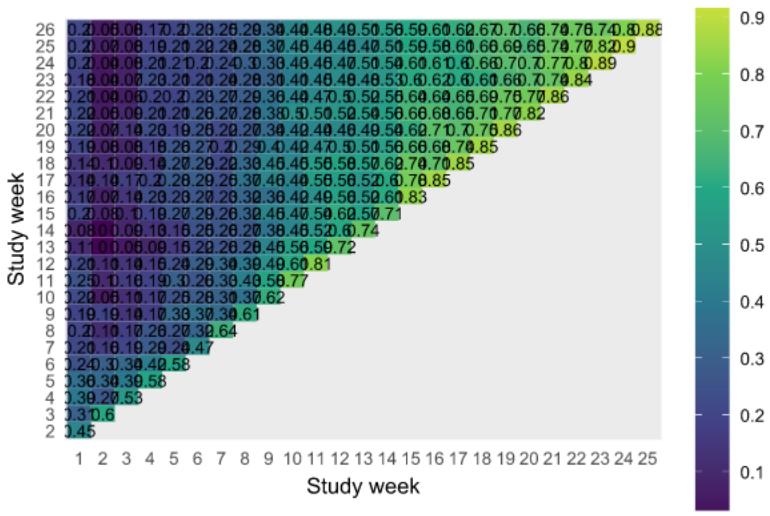

```r
ggsave(filename = here("img", "fig-emp-transition-hosp-weekly.svg"), height = 10, width
= 12)
```

Correlation Plot

Correlation plot

```r
# Create correlation matrix
corr <-
  df_forward_weekly |>
    select(pat_id, week, y) |>
    arrange(week) |>
    mutate(y = as.integer(y)) |>
    pivot_wider(names_from = week, values_from = y) |>
    ungroup() |>
    select(-pat_id) |>
    cor(method="spearman", use="pairwise.complete.obs")

corr |>
  as.data.frame() |>
  rownames_to_column() |>
  pivot_longer(-rowname) |>
  mutate(rowname = as.integer(rowname), name = as.integer(name)) |>
  filter(rowname < name) |> #remove tiles below diagonal, because redundant
  mutate(rowname = as.factor(rowname), name = as.factor(name)) |>
  ggplot() +
    aes(x=rowname, y=name, fill=value) +
    geom_tile() +
    geom_text(mapping=aes(x=rowname, y=name, label=round(value, 2)),
              color="black",
              size = 3) +
    scale_fill_viridis_c(limits=c(-0.01,1), n.breaks = 10) +
    expand_limits(value = c(0, 1)) +
    # cowplot::theme_half_open(font_size=FONT_SIZE,
    #            font_family = "Source Sans Pro") +
    theme(plot.title = element_text(hjust = 0.5),
          panel.grid.major = element_blank(),
          panel.grid.minor = element_blank()) +
    theme(axis.line = element_blank(),
```

```
        axis.ticks = element_blank()) +
  theme(legend.key.height = unit(0.8, "in")) +
  labs(x="Study week",
       y="Study week",
       fill="")
```



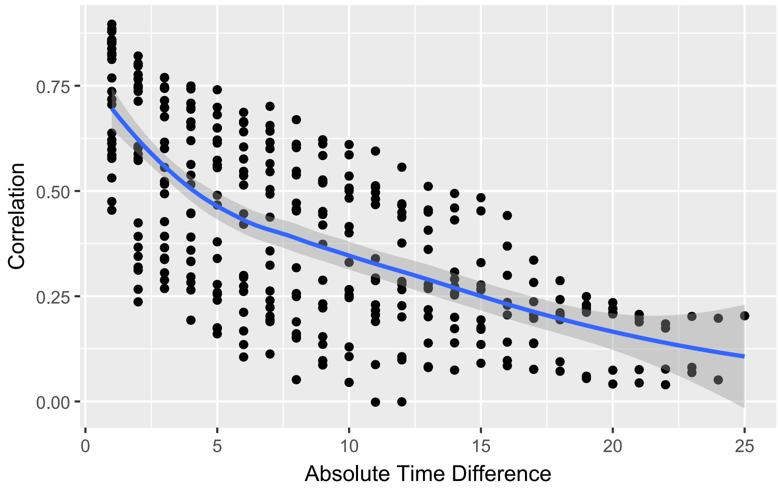

```
ggsave(filename = here("img", "fig-emp-cor-heat-hosp-weekly.svg"), height = 10, width =
11)
```

Variogram

```
plotCorrM(corr)[[2]]
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: The following aesthetics were dropped during statistical transformation:
label.
i This can happen when ggplot fails to infer the correct grouping structure in
  the data.
i Did you forget to specify a `group` aesthetic or to convert a numerical
  variable into a factor?
```




```
ggsave(filename = here("img", "fig-emp-variogram-hosp-weekly.svg"), height = 10, width
= 10)
```

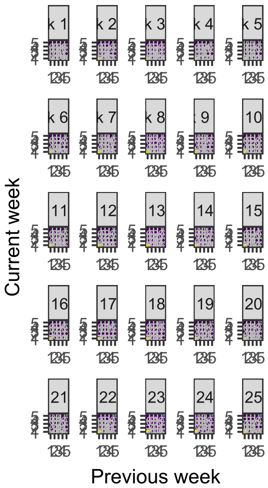
```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: The following aesthetics were dropped during statistical transformation:
label.
i This can happen when ggplot fails to infer the correct grouping structure in
  the data.
i Did you forget to specify a `group` aesthetic or to convert a numerical
  variable into a factor?
```

Transition probabilities

```
max_p <- max(df_transitions$p)

df_transitions |>
  mutate(p = round(p,2)) |>
    mutate(yprev = as.factor(yprev), y = as.factor(y)) |>
    ggplot() +
    aes(x = yprev, y = y, fill = p) +
    geom_tile() +
  geom_text(mapping=aes(x=yprev, y=y, label=p),
              color="gray",
              family = "Source Sans Pro",
              size=2) +
  scale_fill_viridis_c(limits=c(0, max_p), guide = "none") +
    facet_wrap(~week,
                nrow = 5,
                scales = "free",
                labeller = labeller(week = \(x) glue::glue("week {as.integer(x)-1} to
{as.integer(x)}"))) +
    scale_x_discrete(drop=FALSE) +
    scale_y_discrete(drop=FALSE) +
    theme_bw(base_family = "Source Sans Pro",
              base_size = FONT_SIZE) +
   theme(aspect.ratio = 1) +
```

```
    labs(x = "Previous week",
         y = "Current week",
         size = "Proportion")
```

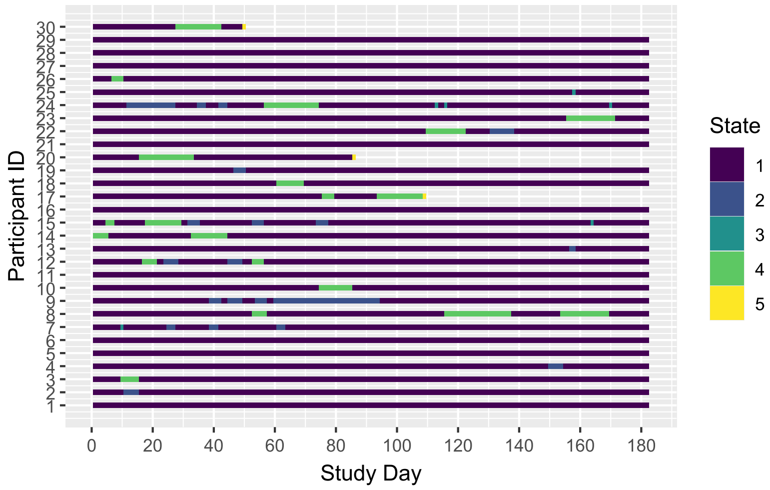



```
ggsave(filename = here("img", "fig-trans-prob-hosp-weekly.svg"), height = 10, width =
10)
```

Daily plots

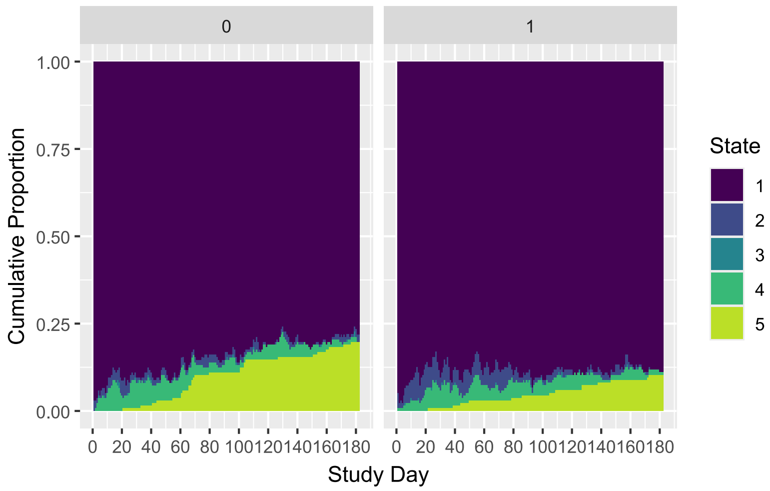
Tile plot

```
df_daily |>
  filter(pat_id %in% ids) |>
  mutate(pat_id = as.integer(factor(pat_id))) |>
  mutate(y = as.ordered(y)) |>
  ggplot() +
  aes(y=pat_id, x=day) +
  geom_tile(mapping = aes(fill = y),
            width=1, height=0.4) +
  scale_fill_viridis_d() +
  scale_x_continuous(n.breaks = 10, limits = c(0.5, 182.5)) +
  scale_y_continuous(breaks = 1:30) +
  labs(x = "Study Day",
       y = "Participant ID",
       fill = "State")
```



Empirical SOPs

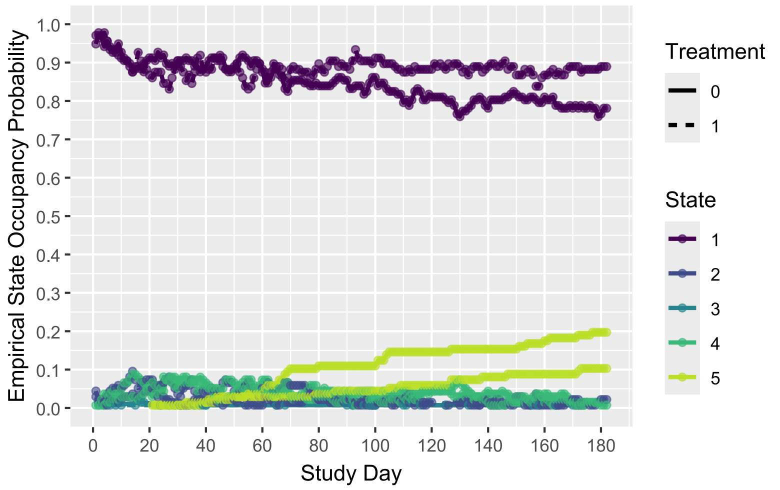
```
df_forward_daily |>
      mutate(y = as.factor(y)) |>
      group_by(day, y, tx) |>
      count() |>
      group_by(day, tx) |>
      mutate(n = n / sum(n)) |>
      ggplot() +
      aes(x = day, y = n, fill=y) +
      geom_col(position="fill", width = 1) +
      scale_x_continuous(limits = c(0.5, 182.5),
                         n.breaks = 10) +
      scale_y_continuous(breaks = seq(0, 1, 0.25)) +
      scale_fill_viridis_d(end = 0.9) +
      coord_cartesian(ylim = c(0, 1)) +
      facet_wrap(~tx, nrow=1) +
      labs(x = "Study Day",
           y = "Cumulative Proportion",
           fill = "State")
```




```
ggsave(filename = here("img", "fig-emp-sop-hosp-daily.svg"), height = 10, width = 12)
```

SOP lines plot

```
df_forward_daily |>
  mutate(y = as.factor(y)) |>
  group_by(day, y, tx) |>
  count() |>
  group_by(day, tx) |>
  mutate(n = n / sum(n)) |>
  ggplot() +
  aes(x = day, y = n, color=y, linetype = tx) +
  geom_line(linewidth = 1) +
  geom_point(alpha=0.6, size=1.5) +
  scale_x_continuous(n.breaks = 10) +
  scale_color_viridis_d(end = 0.9) +
  scale_y_continuous(breaks = seq(0, 1, by=0.1)) +
  coord_cartesian(ylim = c(0,1)) +
  labs(x = "Study Day",
       y = "Empirical State Occupancy Probability",
       color = "State",
       linetype = "Treatment"
  )
```



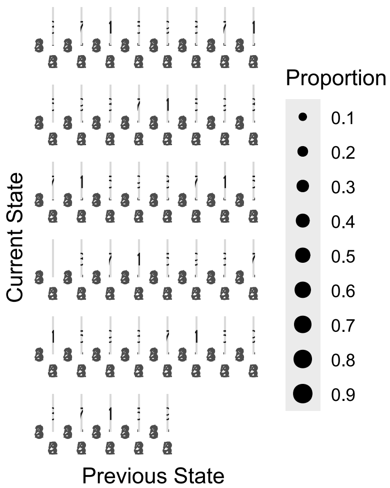

```
ggsave(filename = here("img", "fig-emp-sop-line-hosp-daily.svg"), height = 10, width =
12)
```

Transitions

```
fu <- 182
# Create the transition data.frame for week = x
get_transitions_df <- function(df, x){


  subset <- df[df$day == x, ]
  subtable <- table(subset$y, subset$yprev, dnn = c("y", "yprev")) |>
  as.data.table()

  subtable[, day := x]
  return(subtable)
```

```
}

# Get the transition data frame for each week and combine
# into a single data frame
df_transitions <-
  map(2:fu, ~get_transitions_df(df_forward_daily, .x)) |>
  rbindlist()

# Compute the proportion of transitions relative to the number of
# transitions on each week
df_transitions <-
  df_transitions |>
    mutate(total = sum(N),
           p = N / total, .by=day)
```

```
df_transitions |>
    filter(day%%4 == 0) |>
    mutate(yprev = as.factor(yprev), y = as.factor(y)) |>
    filter(p != 0) |>
    ggplot() +
    aes(x = yprev, y = y, size = p) +
    geom_point() +
    scale_size_area(breaks = seq(0, 1, by = 0.1), max_size = 4) +
    facet_wrap(~day,
               nrow = 6,
               scales = "free",
               labeller = labeller(day = \(x) glue::glue("day {as.integer(x)-1} to
{as.integer(x)}"))) +
    scale_x_discrete(drop=FALSE) +
    scale_y_discrete(drop=FALSE) +
    theme(axis.ticks = element_blank()) +
    theme(aspect.ratio = 0.9) +
    labs(x = "Previous State",
         y = "Current State",
         size = "Proportion")
```

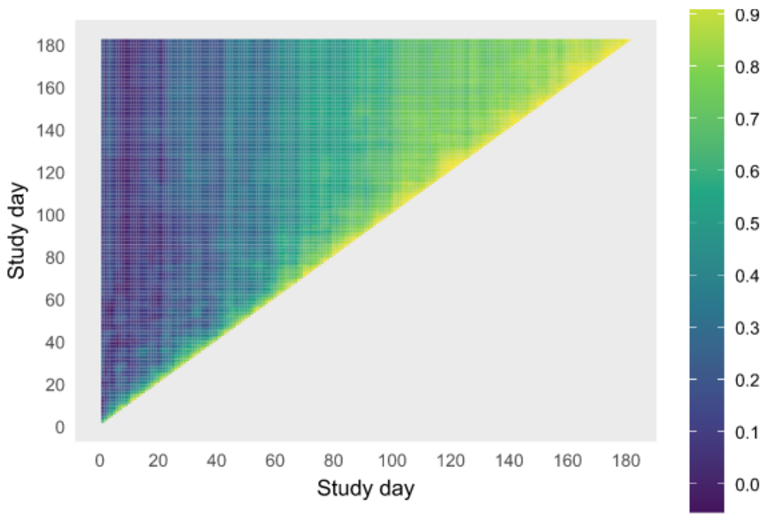


```r
ggsave(filename = here("img", "fig-emp-transition-hosp-daily.svg"), height = 12, width
= 12)
```

Correlation Plot

Correlation plot

```r
# Create correlation matrix
corr <-
  df_forward_daily |>
    select(pat_id, day, y) |>
    arrange(day) |>
    mutate(y = as.integer(y)) |>
    pivot_wider(names_from = day, values_from = y) |>
    ungroup() |>
    select(-pat_id) |>
    cor(method="spearman", use="pairwise.complete.obs")

corr_reduced <-
  df_forward_daily |>
    select(pat_id, day, y) |>
    arrange(day) |>
    mutate(y = as.integer(y)) |>
    filter(day %% 4 == 0) |>
    pivot_wider(names_from = day, values_from = y) |>
    ungroup() |>
    select(-pat_id) |>
    cor(method="spearman", use="pairwise.complete.obs")

corr |>
  as.data.frame() |>
  rownames_to_column() |>
  pivot_longer(-rowname) |>
  mutate(rowname = as.integer(rowname), name = as.integer(name)) |>
  filter(rowname < name) |> #remove tiles below diagonal, because redundant
  # mutate(rowname = as.factor(rowname), name = as.factor(name)) |>
  ggplot() +
    aes(x=rowname, y=name, fill=value) +
    geom_tile() +
    # geom_text(mapping=aes(x=rowname, y=name, label=round(value, 2)),
    #           color="black",
    #           size = 0) +
    scale_fill_viridis_c(limits=c(-0.1,1), n.breaks = 10) +
    expand_limits(value = c(0, 1)) +
    scale_y_continuous(n.breaks = 10) +
    scale_x_continuous(n.breaks = 10) +
    theme(plot.title = element_text(hjust = 0.5),
          panel.grid.major = element_blank(),
          panel.grid.minor = element_blank()) +
    theme(axis.line = element_blank(),
          axis.ticks = element_blank()) +
    theme(legend.key.height = unit(0.8, "in")) +
    labs(x="Study day",
         y="Study day",
         fill="")
```



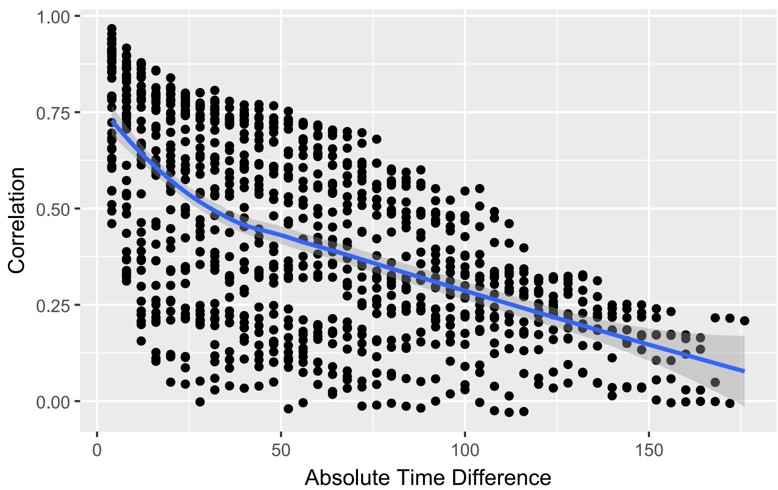

```
ggsave(filename = here("img", "fig-emp-cor-heat-hosp-daily.svg"), height = 10, width =
11)
```

Variogram

```
plotCorrM(corr_reduced)[[2]]
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: The following aesthetics were dropped during statistical transformation:
label.
i This can happen when ggplot fails to infer the correct grouping structure in
  the data.
i Did you forget to specify a `group` aesthetic or to convert a numerical
  variable into a factor?
```



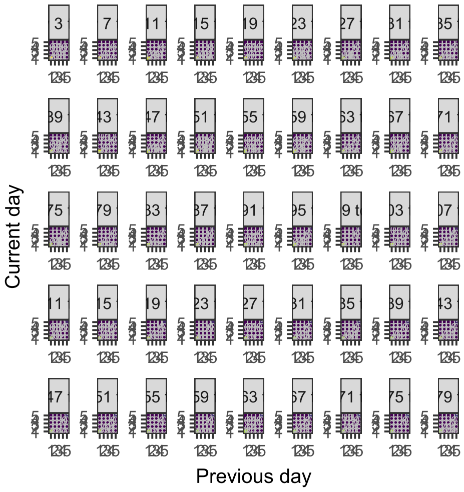

```
ggsave(filename = here("img", "fig-emp-variogram-hosp-daily.svg"), height = 10, width =
10)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: The following aesthetics were dropped during statistical transformation:
label.
i This can happen when ggplot fails to infer the correct grouping structure in
  the data.
i Did you forget to specify a `group` aesthetic or to convert a numerical
  variable into a factor?
```

Transition probabilities

```
max_p <- max(df_transitions$p)

df_transitions |>
  filter(day%%4 == 0) |>
  mutate(p = round(p,2)) |>
  mutate(yprev = as.factor(yprev), y = as.factor(y)) |>
  ggplot() +
  aes(x = yprev, y = y, fill = p) +
  geom_tile() +
  geom_text(mapping=aes(x=yprev, y=y, label=p),
            color="gray",
            family = "Source Sans Pro",
            size=2) +
  scale_fill_viridis_c(limits=c(0, max_p), guide = "none") +
  facet_wrap(~day,
             nrow = 5,
             scales = "free",
             labeller = labeller(day = \(x) glue::glue("day {as.integer(x)-1} to
{as.integer(x)}"))) +
  scale_x_discrete(drop=FALSE) +
  scale_y_discrete(drop=FALSE) +
  theme_bw(base_family = "Source Sans Pro",
           base_size = FONT_SIZE) +
  theme(aspect.ratio = 1) +
  labs(x = "Previous day",
       y = "Current day",
       size = "Proportion")
```




```
ggsave(filename = here("img", "fig-trans-prob-hosp-daily.svg"), height = 8, width = 10)
```